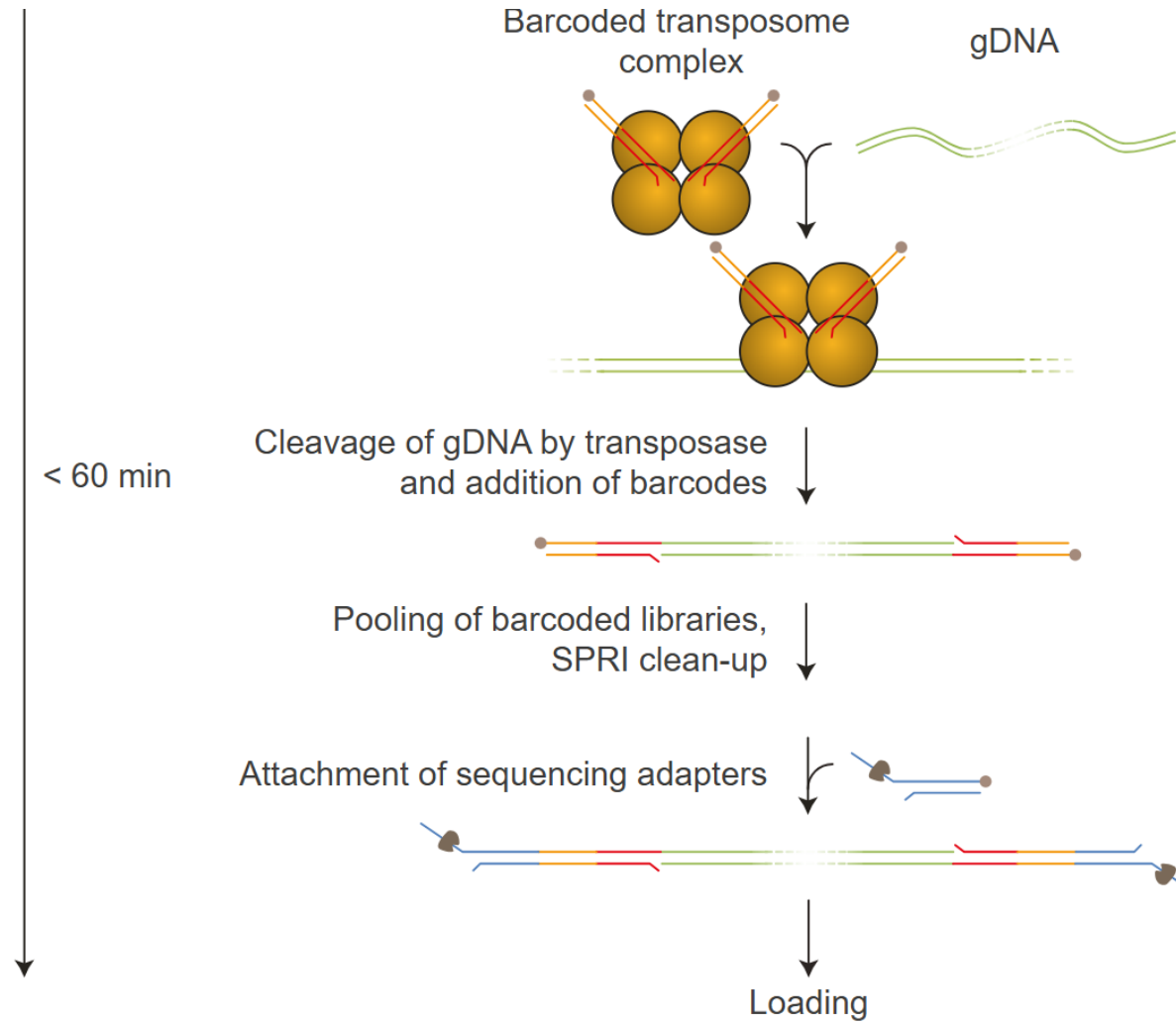




gDNA sequencing using Nanopore: Rapid Barcoding Kit

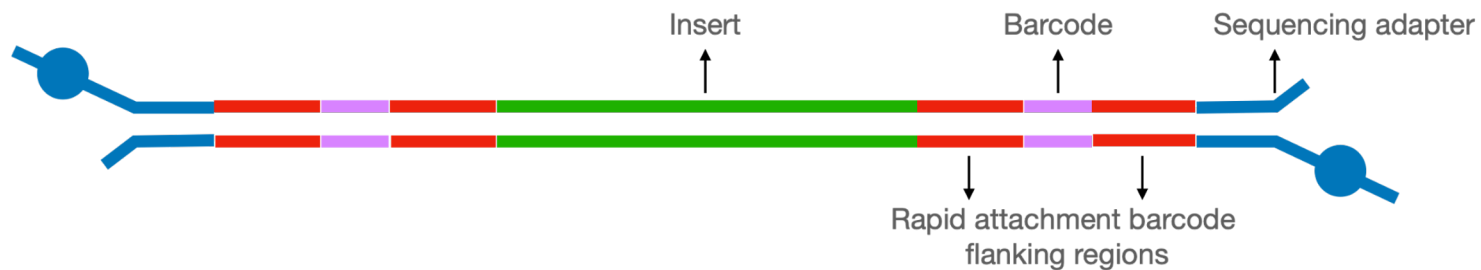
Kanny Diallo
Simon Amoikon

Workflow

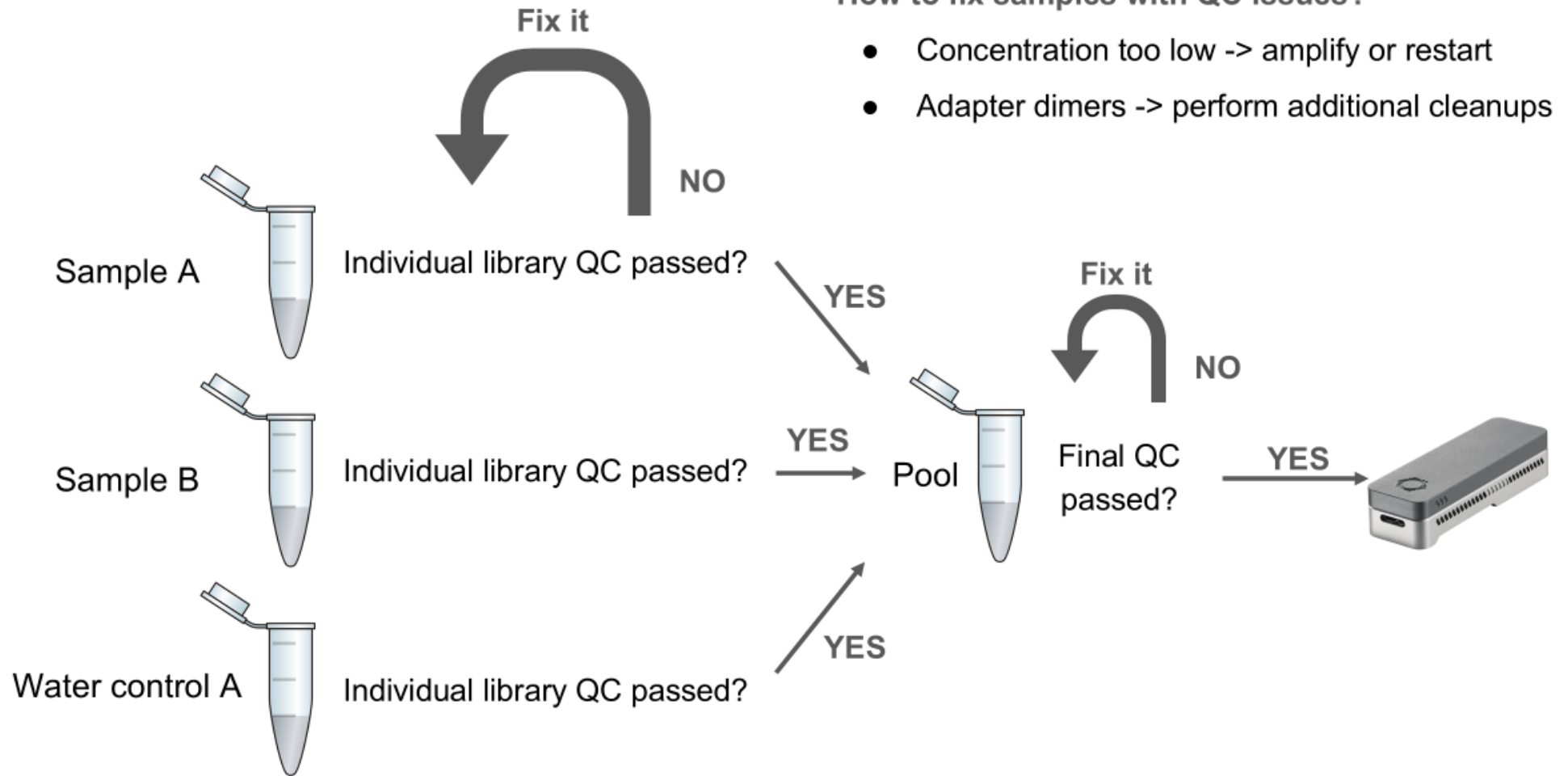


Library Quality Control (QC)

- The purpose of QC is to ensure that there is sufficient and good quality library for sequencing.
- Quantity (concentration)
 - Amount of library is measured using Qubit (or other fluorescence based method)
- Quality (length)
 - Measured using capillary electrophoresis (Tapestation, Bioanalyzer, Fragment Analyzer)
 - If capillary electrophoresis is unavailable, regular gel electrophoresis may be used



Library prep QC



How to fix samples with QC issues?

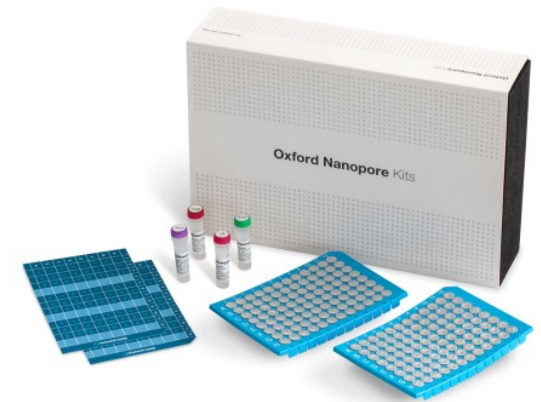
- Concentration too low -> amplify or restart
- Adapter dimers -> perform additional cleanups

Group Organization

- 12 groups total (11 groups of 2; 1 group of 3)
 - ✓ Each group receives pre-aliquoted samples
 - ✓ Each group analyzes 3 pathogens and 1 water control
 - ✓ Each sample gets a unique barcode (assigned before start)

Workflow Overview

1. DNA prep (200 ng per sample)
2. Rapid barcoding reaction
3. Pooling
4. Bead cleanup (AMPure XP)
5. Qubit quantification of each pool
6. Loading final library onto flow cell



Rapid sequencing DNA V14 -
barcoding (SQK-RBK114.96)



GridIon

DNA Preparation

- Transfer 200 ng gDNA (5 μ l of 40 ng/ μ l of gDNA) per sample into PCR tubes
- Adjust to 10 μ l final volume with nuclease-free water
- Pipette-mix 10–15 $\times$

Barcoding Reaction

- Per sample: DNA 10 μ l + Rapid Barcode 1.5 μ l = 11.5 μ l final volume
- Mix thoroughly, spin briefly
- Incubate 30°C for 2 min → 80°C for 2 min, then put on ice

Pooling Scheme (Groups → Pools)

- Each group makes 4 barcoded samples
- 3 groups combine their barcoded samples into one pooled LoBind tube
 - ✓ $(3 \text{ groups} \times 4 \text{ samples} = 12 \text{ barcoded libraries per pool})$
 - ✓ 12 groups → 4 pools total

Bead Cleanup (AMPure XP)

- Vortex beads to resuspend
- Add equal volume of beads to pooled sample,
- mix, incubate 10 min on rotator
- Place on magnet, remove supernatant
- Wash 2× with 80% ethanol, dry briefly (≤ 30 s)
- Elute in EB (15 μ l for ≤ 24 barcodes; scale up for larger pools)

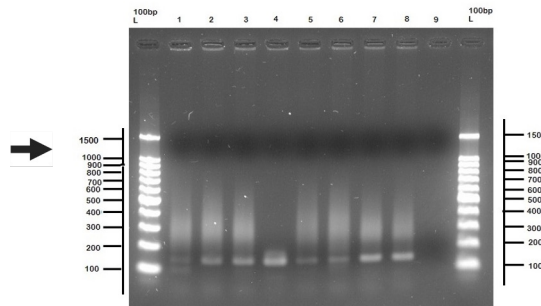
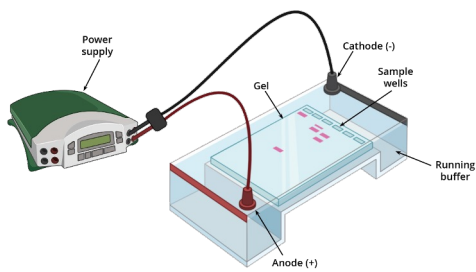


AMPure XP

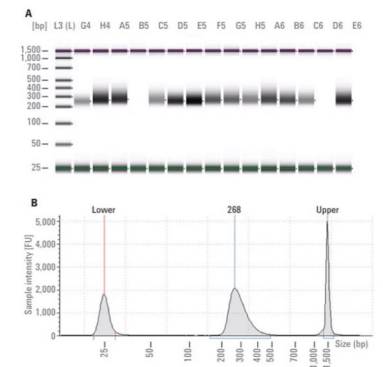


Magnetic separation rack

Library QC Length

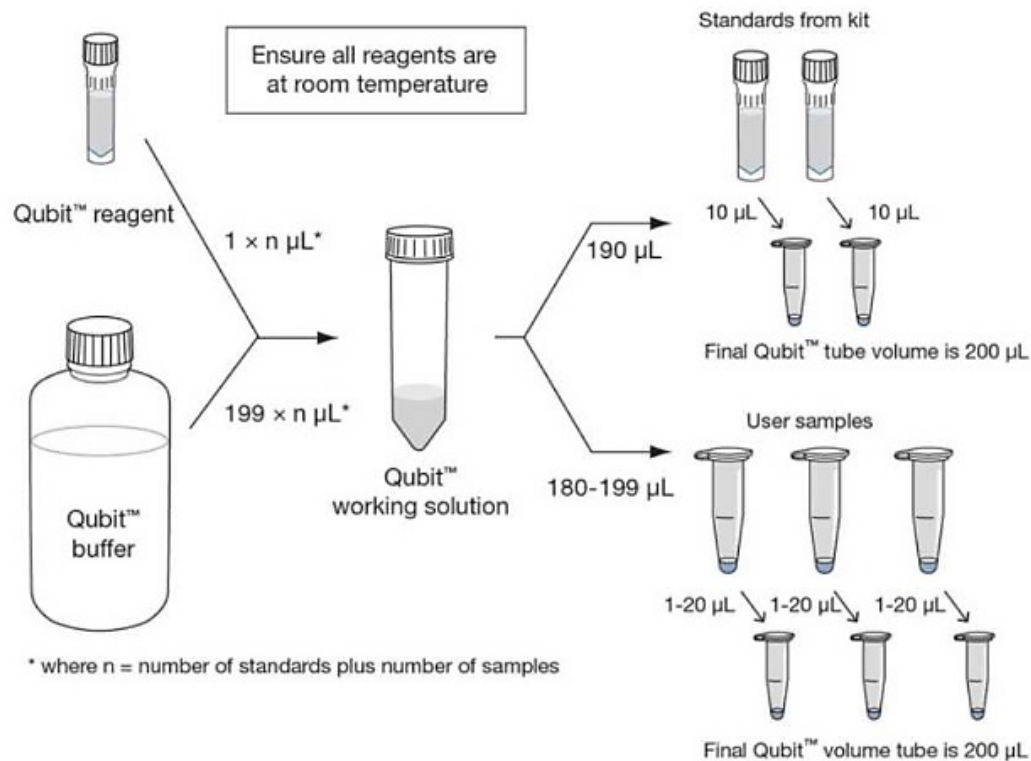


Gel electrophoresis of ladder and samples



Capillary electrophoresis
E.g. Tapestation, Bioanalyzer

Library QC concentration and Final Preparation



- Quantify 1 μL of eluate on Qubit
- Record concentration and volume for each pool
- Transfer 11 μL of library to a clean LoBind tube (per protocol)

Loading on Flow Cell and Notes

- Check flow cell pore count before loading
- Avoid bubbles when pipetting onto the flow cell
- Note which pool(s) are loaded and their barcode ranges



Critical Points

- Ensure correct barcode assignment (pre-assign list)
- Label tubes clearly: Group ID, pathogen, barcode
- Keep sterile technique and change tips between samples

Output fastq file organization

- Reads are automatically demultiplexed during basecalling
- Fastq files are generated in real-time (every 10 minutes in this example), so multiple fastq files are generated for each barcode (sample)

barcode_aliases379472e5.tsv ✓	barcode01	FAY32108_p...5_0.fastq.gz	Aug 13, 2024 at 1:06 PM
fastq_fail	barcode02	FAY32108_p...5_1.fastq.gz	Aug 13, 2024 at 1:16 PM
fastq_pass	barcode03	FAY32108_p...5_2.fastq.gz	Aug 13, 2024 at 1:26 PM
final_summary79472e5.txt ✓	barcode04	FAY32108_p...5_3.fastq.gz	Aug 13, 2024 at 1:36 PM
other_reports	barcode05	FAY32108_p...5_4.fastq.gz	Aug 13, 2024 at 1:46 PM
pod5	barcode06	FAY32108_p...5_5.fastq.gz	Aug 13, 2024 at 1:56 PM
pore_activity...9472e5.csv ✓	barcode07	FAY32108_p...5_6.fastq.gz	Aug 13, 2024 at 2:06 PM
report_FAY3...4b3362.html ✓	barcode08	FAY32108_p...5_7.fastq.gz	Aug 13, 2024 at 2:16 PM
report_FAY3...4b3362.json ✓	barcode09	FAY32108_p...5_8.fastq.gz	Aug 13, 2024 at 2:26 PM
report_FAY3...a4b3362.md	barcode10	FAY32108_p...5_9.fastq.gz	Aug 13, 2024 at 2:36 PM
sample_sheet...a4b3362.csv ✓	barcode11	FAY32108_p...10.fastq.gz	Aug 13, 2024 at 2:46 PM
sequencing...379472e5.txt	barcode12	FAY32108_p...5_11.fastq.gz	Aug 13, 2024 at 2:56 PM
throughput...79472e5.csv ✓	barcode13	FAY32108_p...12.fastq.gz	Aug 13, 2024 at 3:06 PM
	barcode14	FAY32108_p...13.fastq.gz	
	barcode15	FAY32108_p...14.fastq.gz	

Sequencing run QC using the “report_***.html” file

- Check that the amount of data generated matches your expectation (this is a function of library loading concentration, run time, number of pores available for sequencing)
- Typically, for a fresh flow cell with a run that's 48+ hours, ~20 Gbases of data are generated
- If the flow-cell is being re-used and has <800 pores left (like example below), data generated will be lower

^ Run summary

DATA OUTPUT

Estimated bases

3.09 Gb

Reads generated

3.63 M

Estimated N50

860 b

BASECALLING

Reads called

100%

Reads called (min Q score: 9)

3.15 M

480.34 k

Pass

Fail

Bases called (min Q score: 9)

2.44 Gb

347.16 Mb

Pass

Fail

RUN DURATION

Run time

24 hrs 0 mins / 24 hrs 0 mins

Elapsed time



Run limit



Run status

FINISHED · Target runtime has been reached

Read length distribution

- Check that the distribution of read lengths matches expectation (depends on quality of sample going in, whether fragmentation was performed, etc.)
- Distribution should roughly match what was seen in Tapestation (if used), although it's common for read lengths to skew shorter

^ READ LENGTHS · OUTLIERS REMOVED

The read length graph shows the total number of bases vs the read length. The longest 1% of strands are classified as outliers, and excluded to allow focus on the main body of data.

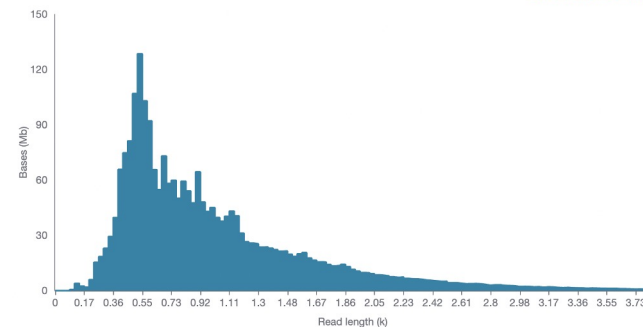
N50*

860 b

% Basecalled

100 %

Basecalled | Estimated



*N50 calculated from basecalled read length histogram.

Cumulative output as a function of time

- Cumulative output should start out linear and then plateau off as pores get exhausted
- >80% of bases should pass the Q-score threshold (usually set to 9)

^ CUMULATIVE OUTPUT

The cumulative output shows the total amount of bases or reads sequenced over time by your device.

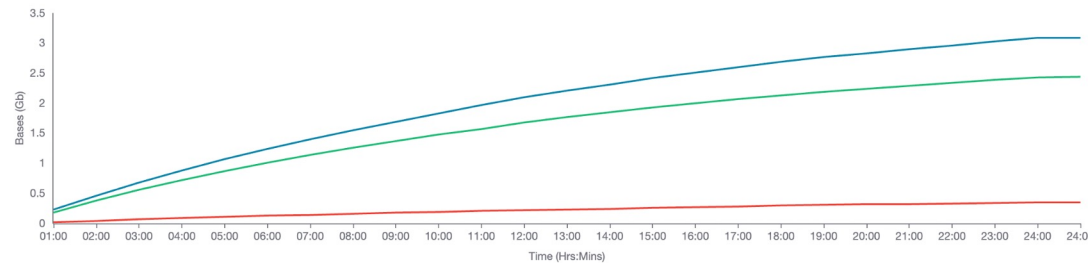
Bases

Legend

— Estimated
Predicted total number of bases, prior to basecalling

— Passed
Bases equal to or above the quality score threshold.

— Failed
Bases below the quality score threshold.



Reads

Legend

— Total
Total number of reads, including passed, failed and skipped.

— Passed
Reads equal to or above the quality score threshold.

— Failed
Reads below the quality score threshold.

— Skipped
Reads that will not be basecalled. Post run basecalling is possible.

Cumulative output as a function of time

- Cumulative output should start out linear and then plateau off as pores get exhausted
- >80% of bases should pass the Q-score threshold (usually set to 9)

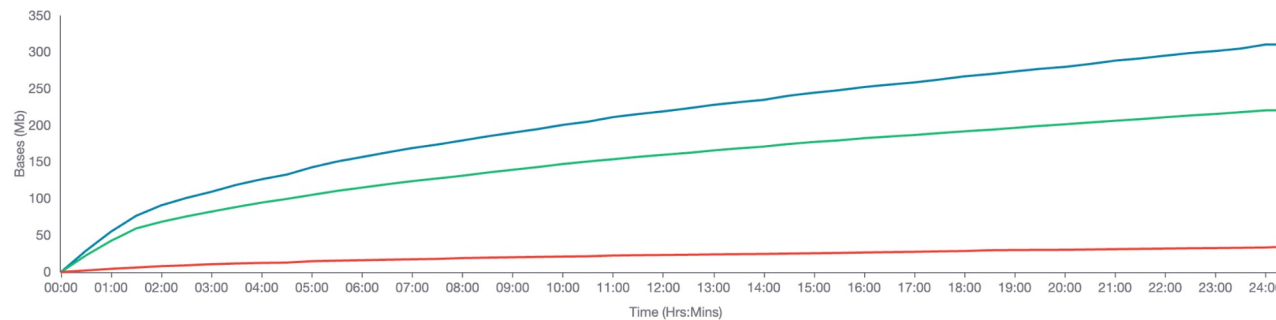
^ CUMULATIVE OUTPUT

The cumulative output shows the total amount of bases or reads sequenced over time by your device.

Bases

Legend

Estimated	Passed	Failed
Predicted total number of bases, prior to basecalling	Bases equal to or above the quality score threshold.	Bases below the quality score threshold.



Cumulative output as a function of time

- Cumulative output should start out linear and then plateau off as pores get exhausted
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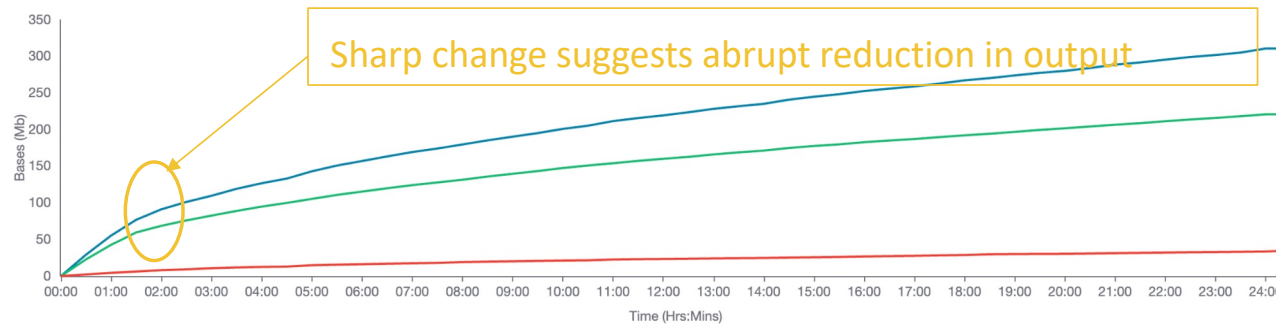
^ CUMULATIVE OUTPUT

The cumulative output shows the total amount of bases or reads sequenced over time by your device.

Bases

Legend

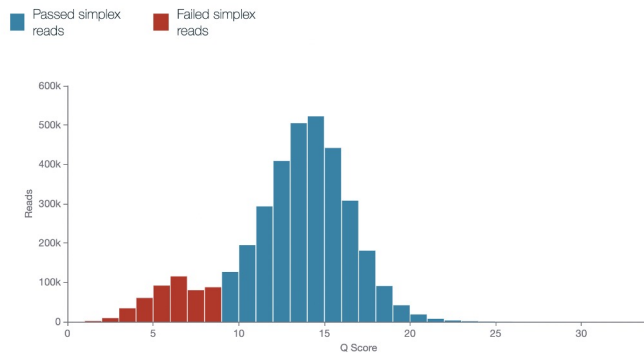
Estimated	Passed	Failed
Predicted total number of bases, prior to basecalling	Bases equal to or above the quality score threshold.	Bases below the quality score threshold.



Q-score

- Q-scores are generally lower than what we get for Illumina; a mean score of 15 is typical
- Ideally, Q-scores will remain steady over time

Q Score histogram



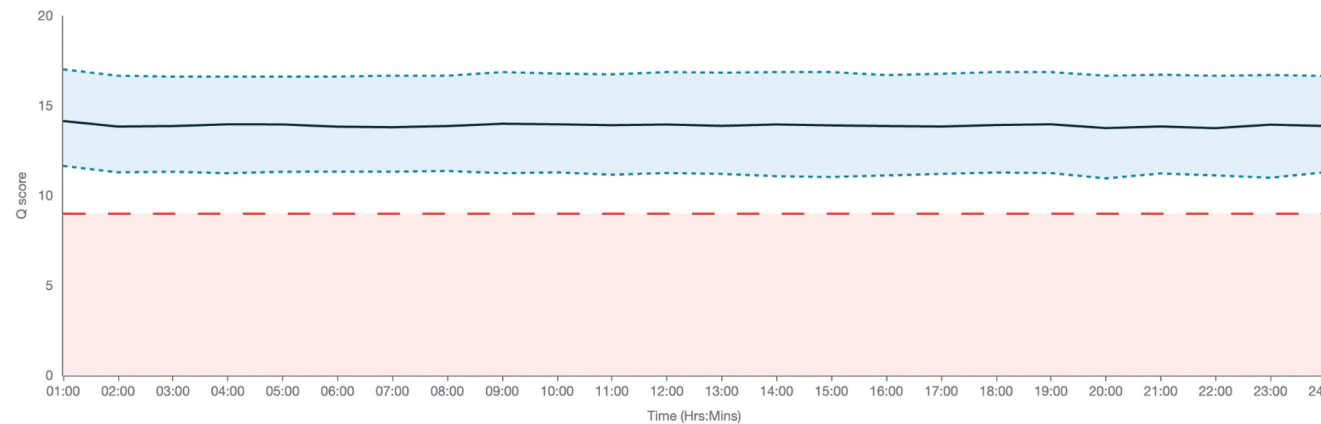
Q Score over time

Legend

— Mode
The most frequent quality score of reads in the run.

Spread
The spread of quality scores, found by calculating full width half maximum.

--- Min. quality score
Minimum quality score to be accepted as a passed read.



Q-score

- Q-scores are generally lower than what we get for Illumina; a mean score of 15 is typical
- Ideally, Q-scores will remain steady over time

^ QUALITY SCORE

The quality score is calculated as basecalling is performed on your device. Reads that fall below the minimum value of 9 will be classified as failed reads. You can alter the accepted minimum quality score in MinKNOW.

Legend

— Mode

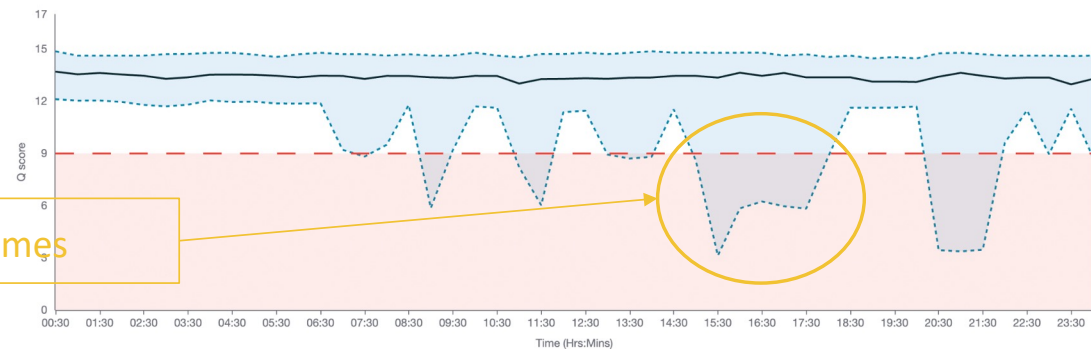
The most frequent quality score of reads in the run.

□ Spread

The spread of quality scores, found by calculating full width half maximum.

--- Min. quality score

Minimum quality score to be accepted as a passed read.



Greater variation (reduction) in Q-score at later times

Pore activity over time

- Sequencing/available pores gradually decrease over time, with slight increases when pore scans are performed

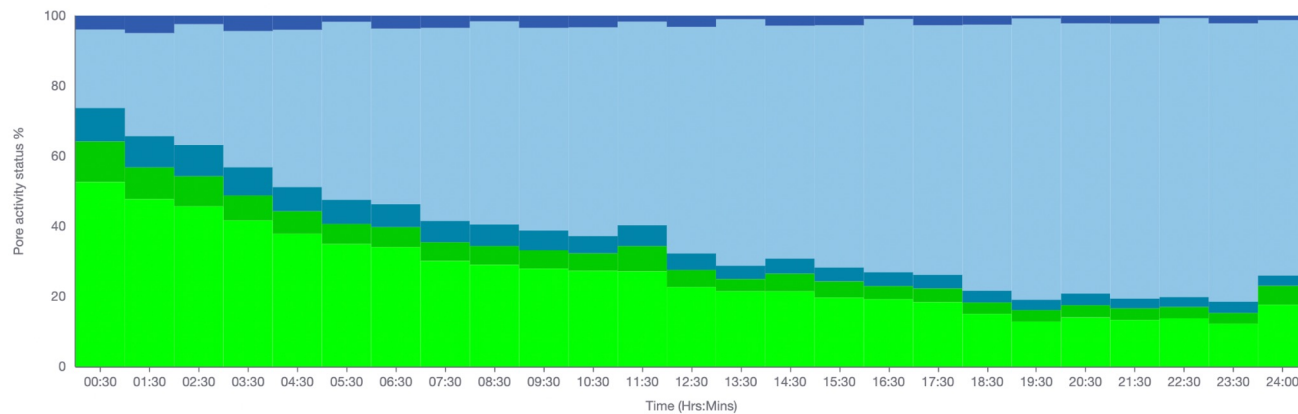
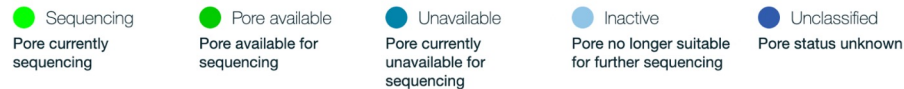
^ Run health

^ PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

☒ Show grouped

Legend



Pore activity over time

- See [here](#) for how to diagnose issues based on pore activity plots

^ PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

Legend

Sequencing Pore currently sequencing	Adapter Pore currently sequencing adapter	Pore available Pore available for sequencing	Unavailable Pore unavailable for sequencing	Active feedback Channel ejecting analyte	No pore No pore detected in channel
Out of range-high Current is positive but unavailable for sequencing	Out of range-low Current is negative but unavailable for sequencing	Multiple Multiple pores detected. Unavailable for sequencing.	Saturated The channel has switched off as current levels exceed hardware limitations	Zero Pore currently unavailable for sequencing	Channel disabled Channel is disabled and awaiting another pore scan
Unclassified Pore status unknown					

Lots of unusual pore statuses

Low starting % of active pores

